

A hybrid genetic algorithm for a home health care routing problem with time window and fuzzy demand

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ABSTRACT

Home Health Care (HHC) companies are widespread in European countries, and aim to serve patients at home to help them recover from illness and injury in a personal environment. Since transportation costs are among the biggest sources of expenditure in company activities, it is of great significance to optimize this in the Home Health Care industry. From the perspective of optimizing the cost of transportation, this paper studies the Vehicle Routing Scheduling problem as it applies to HHC companies. According to a survey of the HHC companies, during the process of delivering medication drugs, the quantity of drugs required for each patient is non-deterministic when the company makes planned routes. This paper considers uncertain demand as a fuzzy variable, which is closer to a potential real life scenario. A Home Health Care Scheduling Problem with fuzzy demand is considered and a fuzzy chance constraint model is designed. We propose a hybrid genetic algorithm integrated with stochastic simulation methods to solve the proposed model. Firstly, the problem is reduced to the classical vehicle routing problem within a time window. Experimental results for Solomon's and Homberger's benchmark instances show that the proposed algorithm performs efficiently. Then other experiments on the fuzzy version model are undertaken with the variable value of the Dispatcher Preference Index (DPI) parameter between [0, 1]. Finally, the influence of DPI on the final objective and the indicators of the problem are discussed using stochastic simulation, and the best value of DPI is obtained. This research will help HHC companies to make appropriate decisions when arranging their vehicle scheduling routes.

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1. Introduction

The World Health Organization has announced that the rate of care-dependent elderly people in Europe will strongly increase in the coming decades (Mankowska, Meisel, & Bierwirth, 2014). In recent years, the health care industry has become one of the largest sectors of the economy in developed countries such as France, Germany, Australia, etc. Due to the issue of global aging and the improvement of treatment conditions, there is no doubt this trend will continue (Golden, Raghavan, & Wasil, 2008).

Home health care (HHC) is a growing medical service in France and other developed countries. This service is provided by the Home Health Care companies, which aim to serve the patients at home to help them recover from illness or injury in a personal environment (Liu, Xie, & Garaix, 2014). Each day a HHC company carries out various logistics activities, including the delivery of drugs or medical instruments from a pharmacy to the patients, and the

transportation of biological samples from patients' home to the laboratory (Liu, Xie, Augusto, & Rodriguez, 2013). A large number of customers is distributed in a given area (such as in a town or a village). Each patient has a service time horizon which is called a time window, and a certain quantity of the drugs is required according to the patient's degree of recovery. Transportation cost is one of the largest forms of expenditure in company activities, so it is of great significance to optimize the vehicle scheduling, which aims to provide the patients a high quality service and reduce costs at the same time.

According to a survey of the HHC companies (Harris, 2015; Liu et al., 2013; Liu et al., 2014), the main operation process of HHC companies can be summarized as follows:

- (1) In the first stage, the HHC company collects information from patients. This information may include: the patient's name, sex, age, address, types of diseases, symptoms, anticipated service time, and other necessary information.
- (2) Then the HHC company designs the planned routes by considering the information collected. At the same time as ensuring the provision of a high quality home health care service (for instance, assisting the patients within their allocated time

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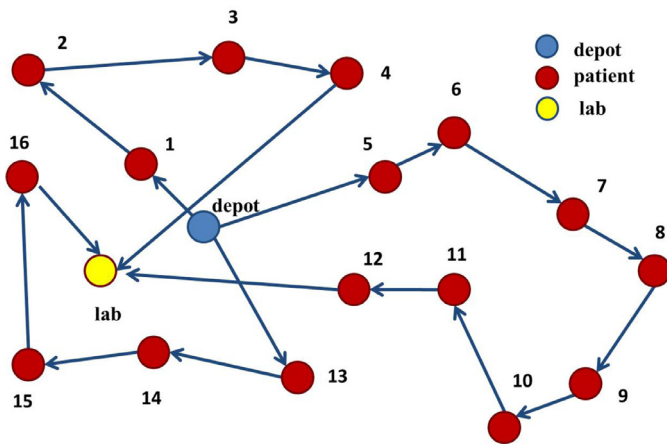


Fig. 1. The specific structure of the HHC supply chain.

horizon and delivering them the right quantity of drugs), the planned route should also be designed to minimize the transportation cost. After all, HHC company is a for-profit organization.

- (3) In the last stage, the nurses are scheduled to visit the patients. Each nurse is assigned to a planned route, and he/she has to carry out all of the service-related activities for the route. This nurse will drive the vehicle to visit the patients one by one according to the designed route. In case of a lack of drugs, the nurse has to go back to the depot, load more drugs into the vehicle and continue to attend to the remaining patients until all the patients are attended to.

Specifically, the HHC company delivers drugs from depot to patients, picks up biological samples from patients to the laboratory, and observes capacity and time window constraints, and this is firmly related to the Vehicle Routing Problem with Time Window (VRPTW) or Vehicle Routing Scheduling Problem (VRSP) (Solomon, 1987). VRPTW is a challenging and difficult problem in the family of Vehicle Routing Problem (VRP) (Toth & Vigo, 2014). In the classical VRPTW, a vehicle starts from a depot, serves a given number of customers in a correct order, and ends at the depot. Each customer is served once and only once. Each vehicle is used no more than once. The objective of the VRPTW is to service all the customers without violating the vehicle capacity constraints with a minimum travel distance or scheduling time.

In our HHC scheduling, each vehicle starts its journey from a depot (i.e. the pharmacy), and terminates its journey at a laboratory. The specific structure of the Home Health Care supply chain can be seen in Fig. 1. This graph consists of one depot, one laboratory and numerous patients (or customers). Additionally, we assume that the drugs have their own volume, so the vehicle capacity must be taken into consideration. However, the samples are vials of blood, or temperature record sheets, which could be assumed to be negligible and will not be considered with respect to the capacity of the vehicle. Our problem is neither like the classical VRP nor like the Open Vehicle Routing problem (OVRP); all the variations of VRP can be seen in the literature (Golden et al., 2008; Pillac, Gendreau, Gu  ret, & Medaglia, 2013; Toth & Vigo, 2014). In the classical VRP, each vehicle needs to return to the depot again, while in the OVRP, each vehicle does not return to the depot after servicing the last customer on a route, but may end at a different location.

Moreover, in the second stage of the operational process, the demand for drugs for each patient is non-deterministic, because the exact quantity of drugs cannot be known unless the nurse arrives at the patient's home. Only by diagnosing the patient for a

certain time (called the working time) can the nurse know the exact quantity of drugs required. However, the HHC company has to design the planned route before the patients are visited. We assume that the demand of each patient is a fuzzy variable.

In this paper we consider the home health care scheduling problem with fuzzy demand, and a Fuzzy Chance Constraint Programming (FCCP) (Liu & Liu, 2002) is constructed based on the fuzzy credibility theory (Liu & Liu, 2002). Due to the capacity constraint of each vehicle, the nurse cannot load unlimited drugs in the vehicle. Unavoidably, sometimes when the nurse reaches the patient, more drugs may be required than planned. In this situation, the nurse has to drive back to the depot, load the drugs into the vehicle, and then drive back to this patient. Then, the nurse will continue to attend to the remaining patients until all the patients are attended to. In this case, the visiting process results in the failure route. There is no doubt that the failure route results in the creation of additional routes, and this may also cause a delay in the service time or can even result in violation of the time window constraint.

We assume that: (1) the vehicles are homogeneous; (2) each vehicle starts from the depot, then visit the patients, and ends at the lab; (3) each vehicle will not be used more than once; (4) each patient cannot be served before the left boundary of its time window; that is, if the nurse arrives at the patient's home in advance, he/she has to wait; (5) the demand of each patient i is a triangular fuzzy variable which can be described as $\tilde{d}_i = (d_{1,i}, d_{2,i}, d_{3,i})$; (6) the distance between i th patient and j th patient is c_{ij} ; (7) each nurse carries out one route in the process of delivery and pickup; in the event that remaining drugs are not sufficient for the next patient, she must return back to the depot, and fill up the capacity of her vehicle.

In this paper, we initially investigate a variety of literature on the VRPTW, HHC and fuzzy demand, propose mathematical formulations of our problem, and then a hybrid genetic algorithm and stochastic simulation method are integrated to solve the proposed model. The rest of the paper is organized as follows: Section 2 introduces the relevant literature. The fuzzy credibility theory is introduced in Section 3, and then the mathematical formulation is built based on the credibility theory in Section 4. Section 5 proposes a hybrid genetic algorithm for solving the problem. Computational experiments are described in Section 6. Section 7 concludes the paper.

2. Literature review

Since our research model is an extension of the classical VRPTW model, we will firstly review the literature on VRPTW, followed by the related research on the Home Health Care and fuzzy demand, respectively.

Over the last fifty years, a considerable amount of research has been focused on the routing optimization problem. The simplest routing problem is the classical traveling salesman problem (TSP). In this problem, a number of cities have to be visited by a salesman, who must drive back to the original city where he started (Ursani, Essam, Cornforth, & Stocker, 2011), and the objective is to optimize the total traveling distance. The Vehicle Routing Problem (VRP) is the m-TSP, where a demand is associated with each city (generally called customer), and every vehicle has a given capacity (El-Sherbeny, 2010).

In recent years, methodological progress and the development of computer technologies have led to increasing academic attention on new variants including more complex objectives and constraints. This trend is simulated by the complex characteristic of real-life VRPS (Lahyani, Khemakhem, & Semet, 2015). This overarching family of the extended problems is often referred to as Rich Vehicle Routing Problems (Lahyani et al., 2015). Many researchers

Table 1
Recent research focus on Rich Vehicle Routing Problem.

Paper	Problem	Solution method
(Ceselli, Righini, & Salani, 2009)	VRP with economies of scale	Column generation algorithm
(Cornillier, Laporte, Boctor, & Renaud, 2009)	Petrol Station Replenishment Problem	Decomposition heuristic
(Derigs, Kurowsky, & Vogel, 2011)	VRP in the air cargo road feeder service	Decomposition heuristic
(Adulyasak, Cordeau, & Jans, 2012)	Electric VRP and recharging stations	Hybrid meta-heuristic
(Hemmelmayr, Doerner, Hartl, & Rath, 2013)	Periodic VRP with intermediate facilities	Hybrid meta-heuristic
(Schneider, Stenger, & Goeke, 2014)	Production routing problem	Adaptive large neighborhood search

have established Rich Vehicle Routing problems by considering the specific characteristics of the supply chains in real life. Some of the recent research on the Rich Vehicle Routing Problem is shown in Table 1.

If we add a time window to each customer, the VRP becomes the Vehicle Routing Problem with Time Window (VRPTW), which has been widely used in the area of bank scheduling, bus scheduling, postal scheduling (Desrochers, Desrosiers, & Solomon, 1992), etc. The CVRP and VRPTW have been proven to be NP-hard problems (Desrochers et al., 1992). In the last few decades, numerous algorithms have been designed to solve the VRPTW, which can be divided into two categories, namely exact methods and heuristic algorithms. The exact methods contain the Branch and Bound, Branch and Cutting, and Lagrange relaxation method (El-Sherbeny, 2010). Although the exact methods can ensure the best solution, it is hard to solve the large scale instances. The earliest researchers have mainly focused on the exact methods. The heuristic algorithms contain the meta-heuristic methods such as Genetic Algorithm (GA), Simulating Annealing (SA), etc. The classical heuristic methods include the savings method, insert method, swap method, the Nearest Neighbor Search, and the hybrid algorithms which are combined by some of the heuristic methods (Baldacci, Mingozzi, & Roberti, 2012). VRPTW has attracted a large number of scholars to study both the heuristic and exact methods. Early surveys of solution techniques can be found in Solomon (1987), Bräysy and Gendreau (2005) and El-Sherbeny (2010).

Despite the increasing importance of the HHC services, few papers deal with HHC problems (Liu et al., 2014). The HHC services have been studied from many perspectives such as the policy, health care facility, logistics (Harris, 2015) etc. Here we investigate the HHC research mainly focusing on its logistic optimization.

Begur, Miller, and Weaver (1997) are among the early scholars to study HHC. They designed a decision support system for HHC in USA, with the help of the savings method and nearest neighbor heuristic. Nickel, Schröder, and Steeg (2012) studied the Home Health Care problem which requires an optimal weekly (or periodical) plan. They proposed a so called master schedule problem, which is an important service process in the HHC. A two-phase method was constructed to solve the model, after which they combined the practical instance with the model. It shows the benefit of using more sophisticated planning methods in the home health care context. Liu et al. (2013) studied the delivery of drugs and medical devices from the Home Care Company's pharmacy to patients' homes, delivery of special drugs from a hospital to patients, and pickup of biometric samples, unused drugs and medical devices from patients. The problem can be considered as a special vehicle routing problem with simultaneous delivery and pickup and time windows, with four types of demand: delivery from depot to patient, delivery from hospital to patient, pickup from patient to depot and pickup from a patient to medical laboratory. Heuristic methods such as GA and TS are proposed to solve the problem.

Additionally, Liu et al. (2014) also studied the periodic home health care logistics based on the classical Periodic Vehicle Routing Problem with Time Window (PVRPTW). Their contribution has been the consideration of the planning horizon which has a set

of possible combinations of visit days. A Tabu Search (TS) method combined with the local search are proposed to solve the model. Recently, Mankowska et al. (2014) have presented a model for the daily planning of health care services carried out at patients' homes by staff members of a home care company. The planning takes into account individual service requirements of the patients, individual qualifications of the staff and possible interdependencies between different service operations. Hiermann, Prandtstetter, Rendl, Puchinger, and Raidl (2015) propose a model which aims to assign home-care staff to customers and determine efficient multimodal tours while considering staff and customer satisfaction. We can conclude that the optimization of the HHC has mainly been undertaken from the perspective of Vehicle Routing scheduling, staff or nurse scheduling and operational planning.

Demand is an important parameter in logistics activities. However, in the previous studies, most research has focused on deterministic demand. In the real world however, it is usually very hard to know the precise demands of customers and thus they are estimated from historical data. Given this aspect of VRPs, a consideration of stochastic vehicle routing problems (SVRP) and fuzzy vehicle routing problems (FVRP) may be useful (Wen & Iwamura, 2008). In the Vehicle Routing Problem with Stochastic Demand, the demand is a stochastic variable, which is decided by probability distribution parameters, but the parameters are often obtained from historical data. Although stochastic models can cater for a variety of cases, they are not sufficient to describe many other situations, where the probability distribution of customers' demands may be unknown or partially known (Wen & Iwamura, 2008). For example, it may be needful to build some new water companies in new regions to service some new customers whose demand can be ascertained neither precisely nor from historical data. However, those demands can be described using non-technical language such as large, small, general, etc. In these cases, fuzzy set theory may do better in dealing with ambiguous information (Wen & Iwamura, 2008). This is the case in the Home Health Care domain, where the quantity of drugs needed by each patient is not deterministic, and has a high relationship with the type of disease, personal health condition, etc. Nonetheless, each type of disease has an upper bound and a lower bound for the quantity of the drugs, and only when the nurse reaches the patient's home, after a check-up, which could involve measuring blood pressure, measuring pulse rate, and talking with the patient personally, can he/she understand the severity of the disease or recovery condition. In this situation, the precise quantity of the drugs can be confirmed. When there is a new patient, the decision-maker would receive an estimation according to the specific type of disease which is easy to describe by fuzzy logic, such as "small penicillin", "general Aspirin". So, it is very appropriate to describe the non-deterministic demand in HHC domain using fuzzy variables. Fuzzy set theory was initialized by Zadeh (1965) and has been widely applied in many real problems. It has been proven to be a useful tool to solve problems with uncertainty. In past decades, many researchers have introduced fuzzy theory into Facility Location Problem (FLP) (Wen & Iwamura, 2008), Location-Routing Problem (LRP) (Mehrjerdi & Nadizadeh, 2013; Zarandi, Hemmati, Davari, & Turksen, 2013), and

Table 2
Recent research focus on the fuzzy credibility theory.

Paper	Problem	Solution method
(Peng & Liu, 2004)	Parallel machine scheduling	Genetic algorithm
(Liu & Li, 2006)	Quadratic assignment problem	Genetic algorithm
(Zheng & Liu, 2006)	Vehicle routing problem	Genetic algorithm
(Yang & Liu, 2007)	Fixed charge solid transportation	Tabu Search
(Zhou & Liu, 2007)	Location-allocation problem	Genetic algorithm
(Liu & Gao, 2009)	Multi-job assignment problem	Genetic algorithm
(Lan, Liu, & Sun, 2009)	Multi-period production	PSO
(Erbao & Mingyong, 2009)	Vehicle routing problem	Differential Evolution
(Li, Protopopescu, Arnold, Zhang, & Gorin, 2009)	Portfolio selection	Simulated Annealing
(Cao & Lai, 2010)	Open vehicle routing problem	Differential Evolution
(Ke & Liu, 2010)	Project scheduling	Genetic algorithm
(Zarandi et al., 2013)	Capacitated Location-routing problem	Simulated Annealing
(Kuo, Zulvia, & Suryadi, 2012)	Vehicle routing problem	PSO
(Sun, Chen, Bi, Wang, & Xiong, 2013)	BOM Product Matching	Quantization Method
(Sadeghi, Sadeghi, & Niaki, 2014)	Hybrid vendor-managed inventory and transportation problem	PSO
(Sarkar & Mahapatra, 2015)	Fuzzy inventory model	Heuristic algorithm

Open Vehicle Routing Problem (OVRP) (Cao & Lai, 2010; Erbao & Mingyong, 2009). More applications on the fuzzy credibility in recent years can be found in Table 2.

By contrast with existing works, our study considers HHC operations from a new perspective. We consider the non-deterministic demand for the drugs, and we assume that the demand for each patient is a fuzzy variable. To the best of our knowledge, our paper is the first to incorporate the vehicle scheduling of visiting patients and the fuzzy demand of each patient.

The main contribution of this paper to the existing body of literature on vehicle routing issues consists of the following: (1) We propose a Home Health Care Routing model based on a survey of the Home Health Care companies. In this model, the vehicle begins its journey from the depot and ends its journey at the laboratory, which is quite different from the classical VRPTW. (2) We assume that the demand for drugs for each patient is a fuzzy variable, which is closer in practice to a real life situation. (3) Stochastic simulation and hybrid genetic algorithm are integrated to solve this problem by focusing on minimizing the total distance. (4) Some new indicators of the service quality are introduced to explain the results.

3. Credibility theory

The concept of fuzzy sets was first discussed by Zadeh (1965) in 1965 with the classical membership function. As a result of his framework, fuzzy sets theory has been applied to many fields, from control theory to artificial intelligence. In order to measure a fuzzy event, the term fuzzy variable was introduced by Kaufmann (1975), while the theory of fuzzy variables to measure possibility was proposed by Zadeh (1978). Although possibility measurement has been used widely, it has no self-dual property. However, a self-dual mathematical property is crucial and necessary in both theory and practice. Liu and Liu (2002) recently developed credibility theory and made the credibility a self-dual measurement. In this section we will first introduce the credibility theory (Liu & Liu, 2002), because it is very important to describe the fuzzy demand.

Let Θ be a non-empty set, and let P be the power set of Θ . Each element in P is called an event, and φ is an empty set. In order to present an axiomatic definition of possibility, it is necessary to assign a number $Pos\{A\}$ to each event, which indicates the possibility that A will occur. In order to ensure that the number $Pos\{A\}$ has a certain mathematical properties that we would intuitively expect possibility to have, we list the following four axioms:

Axiom 3.1 $Pos\{\Theta\} = 1$.

Axiom 3.2 $Pos\{\varphi\} = 0$.

Axiom 3.3 For each $A_i \in P(\Theta)$, $Pos\{\cup_i\} = \sup_i Pos\{A_i\}$.

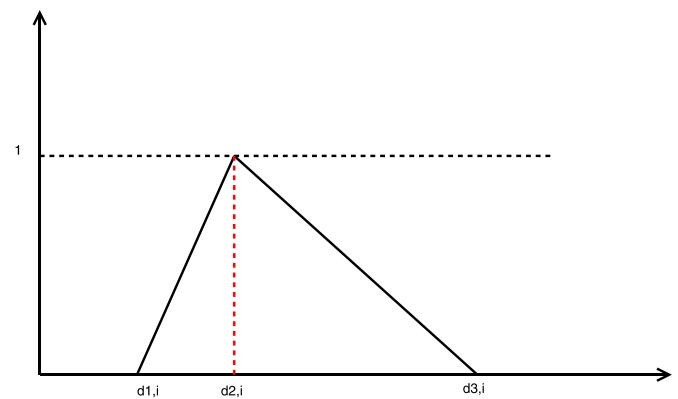


Fig. 2. The description of the fuzzy demand of patient i .

Axiom 3.4 If Θ is a non-empty set, and the set function $Pos\{\cdot\}$, $i = 1, 2, \dots, n$, then for each $A \in P(\Theta)$, $Pos\{A\} = \sup_{(\theta_1, \theta_2, \dots, \theta_n)} Pos\{\theta_1\} \wedge Pos\{\theta_2\} \wedge \dots \wedge Pos\{\theta_n\}$.

The above 4 axioms form the basis of credibility measure theory. Thus we can derive all concepts of credibility theory from them.

Definition 3.1. Let $(\Theta, P(\Theta), Pos)$ be a possibility space, and let A be a set in $P(\Theta)$. Then the necessity of A is defined by $Nec\{A\} = 1 - Pos\{A^c\}$.

Definition 3.2. Let $(\Theta, P(\Theta), Pos)$ be a possibility space, and let A be a set in $P(\Theta)$. Then the credibility of A is defined by $Cr\{A\} = \frac{1}{2}(Pos\{A\} + Nec\{A\})$.

It is clear that the credibility is self-dual which is easily applied to mathematical programming. Additionally, the law of credibility in the theory of fuzzy sets plays a role similar to that played by the law of probability in measurement theory of the ordinary sets.

Let us consider the triangle fuzzy variable $\tilde{d} = (d_1, d_2, d_3)$ as the demand of a given patient, such that d_1 is the left boundary and d_3 is the right boundary, and d_2 is corresponding to a grade of the membership of 1. In practice, the patient's demand will not be less than d_1 or greater than d_3 . The simple description of the fuzzy demand of the patient i can be observed in Fig. 2, and the value of d_1, d_2, d_3 can be subjectively estimated by the dispatchers or the decision-makers according to their working experience.

Based on the definition of possibility, necessity, and credibility, we can derive that:

$$Pos\{\tilde{d} \geq r\} = \begin{cases} 1, & \text{if } r \leq d_2; \\ \frac{d_3 - r}{d_3 - d_2}, & \text{if } d_2 \leq r \leq d_3; \\ 0, & \text{if } r \geq d_3 \end{cases} \quad (1)$$

$$Nec\{\tilde{d} \geq r\} = \begin{cases} 1, & \text{if } r \leq d_1; \\ \frac{d_2 - r}{d_2 - d_1}, & \text{if } d_1 \leq r \leq d_2; \\ 0, & \text{if } r \geq d_2 \end{cases} \quad (2)$$

$$Cr\{\tilde{d} \geq r\} = \begin{cases} 1, & \text{if } r \leq d_1; \\ \frac{2d_2 - d_1 - r}{2(d_2 - d_1)}, & \text{if } d_1 \leq r \leq d_2; \\ \frac{d_3 - r}{2(d_3 - d_2)}, & \text{if } d_2 \leq r \leq d_3; \\ 0, & \text{if } r \geq d_3; \end{cases} \quad (3)$$

4. Mathematical model

In this section, we introduce the mathematical model, which employs fuzzy chance constrained programming. In order to describe the model clearly, firstly, we define some important indicators to describe the vehicle scheduling system, and secondly, we give the fuzzy chance constraints.

4.1. Indicators of the vehicle scheduling system with fuzzy demand

In order to describe and evaluate the vehicle scheduling system with fuzzy demand more clearly, some of the indicators are illustrated before we state the mathematical model.

- (1) **Planned Distance (PD)**: a Home Health Care company needs to design planned scheduling routes before visiting the patients; the expected transportation distance is called the planned distance.
- (2) **Failure Route**: in the process of delivery and pickup, sometimes a nurse arrives at a patient's home and finds that the remaining drugs are insufficient for this patient. He/she has to return back to the depot, replenish the drugs, then drive to this patient again. After having finished attending to this patient, he/she has to continue to attend to the remaining patients until all the patients are served. This situation is called the failure route.
- (3) **Additional Distance (AD)**: In the failure route, the nurse has to return to the depot. After loading the drugs, she also return back to the patient, and this causes the additional distance.
- (4) **Total Distance (TD)**: Total distance is the practical transportation distance, which is the sum of the PD and AD in Vehicle Scheduling System with Fuzzy Demand.
- (5) **Total Delayed Time (TDT)**: In the failure route, sometimes the visiting time of patients may exceed the due time. TDT is the total delay time for all the patients.
- (6) **The Percentage of the Delayed Patients (PDP)**: PDP is the Percentage of the Delayed Patients and accounts for all of the patients.
- (7) **Average Delayed Time (ADT)**: ADT is the mean of the delaying time for the patients for the patients whose visits resulted in a time delay.

Remark 1. These indicators are designed for the non-deterministic vehicle scheduling system. Specifically for the deterministic Vehicle Scheduling System, there are no failure routes, so the AD equals 0, TD equals PD, and other indicator are equal to 0.

Remark 2. For a HHC company, the objective is to provide a high quality, timely service for the patients, as well as to cut transportation costs. The indicators PD, AD, TD show the transportation cost, while TDT, PDP, and ADT reflect the service quality. PD, AD and TD are adopted from the literature (Cao & Lai, 2010; Erbao & Mingyong, 2009; Zarandi, Hemmati, & Davari, 2011; Zarandi et al., 2013), while TDT, PDP, and ADT are newly designed for the vehicle scheduling problem in this paper.

4.2. Description of fuzzy chance constraints

In the deterministic VRP, it is straightforward to describe the capacity constraints: the total demand of the whole route should not exceed the vehicle capacity. However, in the VRPTWFD, the capacity constraints become more complex than the deterministic ones for the uncertain demand. Now, we have to consider the relationship between the fuzzy demand and the capacity of the vehicle (Mehrjerdi & Nadizadeh, 2013; Mousavi & Niaki, 2013).

Indeed, after serving the j th patient, the remaining capacity is changed and it becomes a fuzzy variable named $\tilde{Q}_j$, where

$$\tilde{Q}_j = q - \sum_{i=1}^j \tilde{d}_i = \left(q - \sum_{i=1}^j d_{3,i}, q - \sum_{i=1}^j d_{2,i}, q - \sum_{i=1}^j d_{1,i} \right) = (Q_{1,j}, Q_{2,j}, Q_{3,j}).$$

In the deterministic model, if the remaining capacity of the vehicle is higher than the customer's demand, this vehicle has the chance to serve this customer. However, facing a fuzzy variable of demand and remaining capacity, how can it be decided whether the vehicle should continue visiting the $(j+1)$ th patient or go to the lab directly? It should be compared with the demand of the $(j+1)$ th patient, which is also a fuzzy variable. Based on the credibility we can derive Eqs. (4) and (5) as follows:

$$\begin{aligned} Cr &= Cr\{\tilde{d}_{j+1} \leq \tilde{Q}_j\} \\ &= Cr\{(d_{1,j+1} - Q_{3,j}, d_{2,j+1} - Q_{2,j}, d_{3,j+1} - Q_{1,j}) \leq 0\} \end{aligned} \quad (4)$$

$$Cr = Cr\{\tilde{d}_{j+1} \leq \tilde{Q}_j\} = \begin{cases} 0, & d_{1,j+1} \geq Q_{3,j} \\ \frac{Q_{3,j} - d_{1,j+1}}{2(Q_{3,j} - d_{1,j+1} + d_{2,j+1} - Q_{2,j})}, & d_{1,j+1} \leq Q_{3,j}, d_{2,j+1} \geq Q_{2,j} \\ \frac{d_{3,j+1} - Q_{1,j} - 2(d_{2,j+1} - Q_{2,j})}{2(Q_{2,j} - d_{2,j+1} + d_{3,j+1} - Q_{1,j})}, & d_{2,j+1} \leq p_{2,j}, d_{3,j+1} \geq Q_{1,j} \\ 1, & d_{3,j+1} \leq Q_{1,j} \end{cases} \quad (5)$$

There is no doubt that if the quantity of remaining drugs is very high, and the demand of the next patient is very low, then the chance of the vehicle of being able to provide the next patient's service becomes greater.

We will describe the preference index by Cr , which denotes the magnitude of our preference to drive the vehicle to the next patient after it has served the current patient according to formulation (5). Note that $Cr \in [0, 1]$. When $Cr = 0$, we declare that the vehicle does not have the capacity to serve the next patient and it should terminate service at the current patient and return to the depot to replenish drugs. When $Cr = 1$, we can be completely sure that the vehicle should serve the next patient. However, in most cases, Cr is neither 0 nor 1, but $Cr \in (0, 1)$.

To describe Cr in a convenient way, let us introduce the dispatcher preference index DPI , where $DPI \in [0, 1]$. Note that DPI expresses the dispatcher's attitude toward risk. When the dispatcher is not a risk-averse, he/she will choose lower values of parameter

DPI, which indicates that the dispatcher prefers to use the vehicle available capacity as much as possible, although there is an increase in the number of cases in which the vehicle arrives at the next customer's home and is not able to carry out planned service due to small available capacity. On the other hand, when the dispatcher is a risk-averse, he will choose greater DPI, this may result in a less complete utilization of vehicle capacity along the planned routes and less additional distance to cover due to failures (Cao & Lai, 2010; Ghaffari-Nasab, Ahari, & Ghazanfari, 2013; Mousavi & Niaki, 2013; Sarkar & Mahapatra, 2015).

4.3. MIP formulation

In this section, we will describe the mathematical model in detail.

First of all, we give the mathematical notations of the model:

- V : the set of the vehicles.
- N : the set of all the vertex in the graph, including the depot, patients, and laboratory.
- $\tilde{d}_i = (d_{1,i}, d_{2,i}, d_{3,i})$: the fuzzy demand of the patient i .
- C : the set of the patients.
- $i, j = 0, 1, 2, \dots, n+1$ is the index of all the nodes in the graph. In particular, 0 stands for the index of the depot, $n+1$ stands for the index of the lab, and the others are for the patients.
- $k = 1, 2, \dots, K$ stands for the index of the vehicle.
- q : the capacity of the vehicle.
- w_i : the working time for the doctor/nurse to service the i th patient
- t_{ij} : the traveling time from i to j
- e_i : the earliest service time for the i th patient
- l_i : the latest service time for the i th patient
- M : a positive large number, this parameter is introduced to construct the time window constraints.
- u_i : an artificial variable which is used to construct the sub-tour constraint.
- f_1 : the additional distance which can be estimated by the stochastic simulation.

We will now describe the mathematical model with the most common used 3-index method. First, we give the 2 decision variables as follows:

$$x_{ijk} = \begin{cases} 1 & \text{if the } k\text{th vehicle travels from patient } i \text{ to patient } j; \\ 0 & \text{otherwise.} \end{cases}$$

$s_{ik} \in \mathcal{R}$ denotes the time at which vehicle k begins to service the patient i . In the event that the given vehicle k does not service patient i , s_{ik} does not mean anything. The mathematical model can be stated as follows:

$$\min f = \sum_{k \in V} \sum_{i \in N} \sum_{j \in N} c_{ij} x_{ijk} + f_1 \quad (6)$$

subject to,

$$\sum_{k \in V} \sum_{j \in N} x_{ijk} = 1 \quad \forall i \in C \quad (7)$$

$$\text{Cr} \left(\sum_{i \in C} \tilde{d}_i \sum_{j \in N} x_{ijk} - q \geq 0 \right) \geq \text{DPI} \quad \forall k \in V \quad (8)$$

$$\sum_{j \in N} x_{0jk} = 1, \quad \forall k \in V \quad (9)$$

$$\sum_{i \in N} x_{ihk} - \sum_{j \in N} x_{hjk} = 0, \quad \forall h \in C, \forall k \in V \quad (10)$$

$$\sum_{i \in N} x_{i(n+1)k} = 1, \quad \forall k \in V \quad (11)$$

$$s_{ik} + t_{ij} + w_i - M(1 - x_{ijk}) \leq s_{jk}, \quad \forall i, j \in N, \forall k \in V \quad (12)$$

$$e_i \leq s_{ik} \leq l_i, \quad \forall i \in N, \forall k \in V \quad (13)$$

$$x_{ijk} \in \{0, 1\}, \forall i, j \in N, \forall k \in V \quad (14)$$

$$u_i - u_j + q * \sum_{k=1}^K x_{ijk} \leq q - d_{1,j}, \forall i, j = 0, 1, \dots, N+1, i \neq j; \quad (15)$$

The objective function (6) is the total distance which includes the planned distance and the additional distance f_1 , where the additional distance is caused by failure routes. Constraints (7) indicate that each customer is visited once only, and constraints (8) mean no vehicles are loaded with more than their capacity under the fuzzy credibility theory. Constraints (9) mean each vehicle starts from the depot. Constraints (10) mean that the vehicle visits the patient and then leaves the patient. Constraints (11) mean that the vehicle ends at the laboratory. Constraints (12) state that each vehicle k cannot arrive at j before $s_{ik} + w_i + t_{ij}$, because the nurse needs the working time w_i for patient i , as well as to travel from i to j , here M is a large scalar. Constraints (13) guarantee that time windows are observed, and constraints (14) are the integrability constraints. Constraints (15) are used to eliminate subtours from the solution.

To the best of our knowledge, this model has not been studied before. Here, we name this model as Vehicle Routing Problem with Time Window and Fuzzy Demand (VRPTWFD).

Remark 1. In the experiment section, sufficient available vehicles are given to enable some of the unused vehicles to be assigned as empty routes. In this situation, each unused vehicle is assumed to start from the depot and its journey ends at laboratory without visiting patients. In order to prevent the empty route from adding to the planned distance, we assume that $c_{0(n+1)} = 0$, even if the laboratory and the depot are not in the same position.

Remark 2. If $d_{1,i} = d_{2,i} = d_{3,i}$, this problem becomes a deterministic model, and in this situation, the additional distance $f_1 = 0$.

The mathematical model is a typical Fuzzy Chance Constraint Programming (FCCP) model, which is a young branch of uncertain programming. Moreover, if we assume the fuzzy variable as a deterministic variable, the VRPTWFD is reduced to a VRPTW. Since VRPTW has been proven to be NP-hard problem, it comes as no great surprise that our model VRPTWFD is also an NP-hard problem with fuzzy chance constraints, which is a more difficult problem. To the best of our knowledge, no researchers have produced similar work.

Inspired by previous work (Tan, Lee, Zhu, & Ou, 2001; Ursani et al., 2011), heuristic algorithms can be employed to solve the VRPTW, and stochastic simulation can be applied to simulate the fuzzy chance constraints. So we develop a hybrid algorithm which integrates the hybrid genetic algorithm with the stochastic simulation method to solve the proposed model.

5. Hybrid genetic algorithm

In this section, a hybrid genetic algorithm (HGA) is proposed to solve the Vehicle Routing Problem with Time Window and Fuzzy Demand. Generally speaking, in the first stage, we apply the route construction method to generate initial feasible solutions, then hybrid genetic algorithm is employed to improve the initial solution. To accelerate the convergence, elitism selection and local search are employed, while to make the solution escape from the local optima in advance, mutation operators and crossover operators are designed. Considering that our problem is a Fuzzy Chance Constraint Programming (FCCP) problem, Stochastic Simulation method is designed to evaluate the each solution candidate.

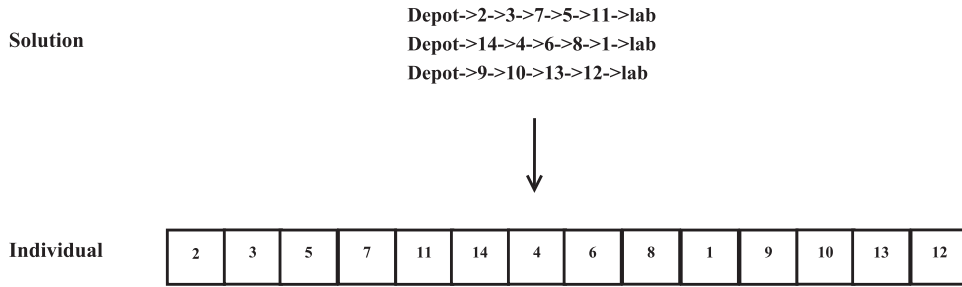


Fig. 3. The first encoding method.

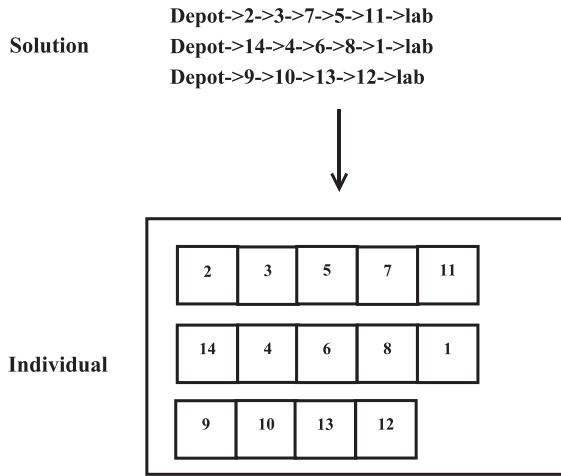


Fig. 4. The representation of an individual.

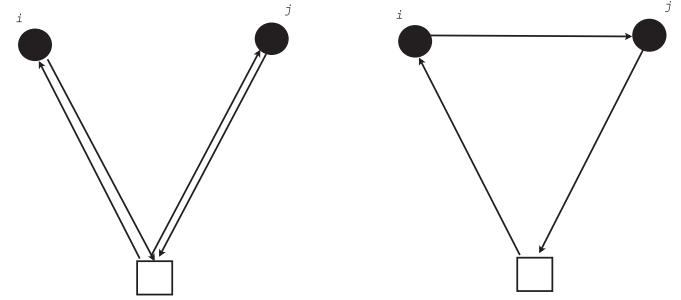


Fig. 5. The savings method.

length of each route may be different. The strength of this encoding method is that each individual does not need to be decoded and encoded in the optimization process.

5.1. Individual representation

In the Genetic Algorithm, an individual represents a solution. Genetic algorithms have been applied to solve the VRPTW by many scholars, while there are just few common ways to encode the individuals. The most common used are two types.

The first encoding method can be seen in Fig. 3. In this instance, the representation of a solution is an integer string of length N , where N is the number of customers. Each gene in the string is an integer node number assigned to the corresponding customer. The sequence of the genes in the chromosome is the order of visiting the customers. It simply links the last customer route i with the first customer visited in the route $i + 1$ to form the string of all the routes involved, but does not put any bit in the string to indicate the end of the routes. There is a chance that it may not reproduce the original routes after decoding. Another problem is that it could not be decoded effectively, especially since there are many nodes whose time window is wide (Tan et al., 2001; Ursani et al., 2011).

Another commonly used encoding method, similar to the first one, employs a string "0" to indicate the end of each route (Tan et al., 2001). This is efficient in decoding the chromosome, because the bit "0" gives the information to cut the string into the routes. Sometimes, the number of vehicles of each solution may not be the same, consequently, the length of each chromosome may be different. This encoding method is less appropriate to design the crossover and mutation operators.

To avoid the problem described above, we propose a new encoding method which employs a list table to describe an individual. Each individual includes many routes.

As shown in Fig. 4, each solution can easily be transformed into a list table, which represents an individual corresponding to a feasible solution. In the list table, each row stands for a route, and the

5.2. Initial population

5.2.1. Route construction method

For the broad application of the VRPTW, many heuristic methods have been proposed to provide solutions. One excellent method is an extension of the savings heuristic of Clarke and Wright (1964). The savings method was first proposed by Clarke and Wright (Solomon, 1987), and is used for solving the capacity vehicle routing problem. The savings method is represented in Fig. 5. The savings method is the best-known routing construction heuristic method. In this method, every patient i, j is serviced by a separate route at first. Combining the two routes results in a cost saving of $S_{ij} = d_{i0} + d_{0j} - d_{ij}$. Clarke and Wright (1964) select arc (i, j) , linking patients i and j with maximum S_{ij} , subject to the requirement that the combined route is feasible. With this conservation, the route combination is applied iteratively.

Solomon (1987) has proposed three sequential insertion heuristics which are named "I1", "I2", and "I3". I1 has been proven to be the most successful one. A route is first initialized with a "seed" patient and the remaining unrouted patients are added into this route until it is full, with the consideration of the time window constraints and the capacity constraint.

After initializing the seed patient, the method uses two defined criteria, $c_1(i, u, j)$ and $c_2(i, u, j)$ to select the patient u for inserting between the adjacent patients i and j in the current partial route. Let $(i_0, i_1, i_2, \dots, i_m)$ be the current route, with i_0, i_m representing the depot and laboratory respectively. For each unrouted patient u , we first compute the best feasible insertion cost on the route as:

$$c_1(i(u), u, j(u)) = \min_{\rho=1, \dots, m} c_1(i_{\rho-1}, u, i_{\rho}) \quad (16)$$

Then, the best patient is u^* to be inserted in the route for:

$$c_2(i(u^*), u^*, j(u^*)) = \min_u \{c_2(i(u), u, j(u)) \mid u \text{ is unrouted and the route is feasible}\}. \quad (17)$$

Patient u^* is then inserted into the route between $i(u^*)$ and $j(u^*)$. When there are no more feasible insertions, a new route is started, unless all the patients have already been routed. $c_1(i, u, j)$ can be calculated as:

$$c_1(i, u, j) = \alpha_1 c_{11}(i, u, j) + \alpha_2 c_{12}(i, u, j), \quad (18)$$

where

$$\alpha_1 + \alpha_2 = 1, \alpha_1 \geq 0, \alpha_2 \geq 0, \quad (19)$$

$$c_{11}(i, u, j) = d_{iu} + d_{uj} - \mu d_{ij}, \mu \geq 0, \quad (20)$$

$$c_{12}(i, u, j) = b_{ju} - b_j, \quad (21)$$

and d_{iu} , d_{uj} and d_{ij} are the distances between i and u , u and j , i and j . Parameter μ controls the savings in distance, b_j represents the beginning service time of patients j , b_{ju} represents the new beginning service time at patient j , in the situation that u is inserted on the route. The criterion $c_2(i, u, j)$ can be calculated as follows:

$$c_2(i, u, j) = \lambda d_{0u} - c_1(i, u, j), \lambda \geq 0. \quad (22)$$

Parameter μ indicates the extent to which the best insertion place for unrouted patient depends on its distance from the depot, and the extent to which the best place also depends on the extra time required to visit the patient by the current vehicle.

When applying this insertion method to the fuzzy demand model, we introduce the formulation (5) to judge whether the next node should be inserted on the route.

5.2.2. Creation of the initial population

An initial population is built such that each individual is a feasible solution. In order to generate a diversity of initial individuals, two methods are used at this stage. Few of the initial individuals are generated by using the Route Construction method which we have introduced in Section 5.2.1; there is no doubt that these individuals showed a high quality. The remaining individuals are generated by selecting the customers in a random manner and inserting them in an existing route if a feasible insert position can be found, otherwise a new route is created, and this process is repeated until all the patients are inserted to the routes.

A parameter named *Rand_Ratio* is defined to determine the percentage of these two types of initial individuals. *Rand_Ratio* is the percentage of the individuals which are generated by random insertion. In our experiments the *Rand_Ratio* = 0.9.

5.3. Fitness value evaluation

As mentioned above, the demand of each patient is a triangular fuzzy number, so it cannot be directly considered as a deterministic number as in the usual algorithms to solve the VRPTW.

Regarding the simulation of the phases of the operation process of the HHC company and related works, we derive an approximate estimation about the indicators we have proposed in Section 4.1 by using a stochastic simulation algorithm. The detailed steps of stochastic simulation (Erbao & Mingyong, 2009; Mousavi & Niaki, 2013) can be described as follows:

Step 1: For each patient, estimate the proposed indicators in Section 4.1 by simulating 'actual' demand. The 'actual' demands were generated by the following process:

- (1) randomly generate a real number x in the interval between the left and right boundaries of the triangular fuzzy number representing demand of the patient, and compute its membership μ ;
- (2) generate a random number a , $a \in [0, 1]$;
- (3) compare a with μ , if $a \leq \mu$, then "actual" demand at the patient is adopted as being equal to x ; in the opposite case, if $a < \mu$, it is not accepted that demand of the patient equals x . In this case, random numbers x and a are generated again and again until random numbers x and a satisfy the relation $a \leq \mu$;
- (4) repeat step (1)–(3), and terminate the process when each patient has a simulated "actual" demand quantity.

Step 2: Move along the route designed by hybrid genetic algorithm, calculate the Additional Distance (AD), Planned Distance (PD), Total Delayed Time (TDT), Percentage of the delayed patients (PDP), and Average of the Delayed Time (ADT) due to route failures in terms of the "actual" demand.

Step 3: Repeat Step 1 and Step 2 for M times.

Step 4: Compute f_1 which is the average value of additional distances obtained by M times simulation.

In the fuzzy demand model, the objective is to minimize the total distance, so the fitness value is $f = \frac{1}{AD+PD}$. Significantly, if $d_{1,i} = d_{2,i} = d_{3,i}$, this problem becomes a deterministic model. In this situation, the additional distance $f_1 = 0$, can be described as $f = \frac{1}{PD}$.

5.4. Selection

Selection is an important part of an evolutionary algorithm. Without selection directing the algorithm towards fitter solutions there would be no progress. Selection must favor fitter candidates over weaker candidates, but beyond that there are no fixed rules.

In the evolution process, some of the good individuals may lost due to the crossover and mutation. This is not in favor of the convergence of the population. In order to overcome this disadvantage, we employ the elitism selection operator. Specifically, in each generation, we remain a few good individuals which are not taken part in the crossover, mutation and local search process.

5.5. Crossover

Crossover is a very important way to reproduce the offspring which can increase the diversity of the population. According to our encoding method, each integer gene appears only once in the chromosome. If we use the traditional crossover method, it may unavoidably generate invalid offsprings which have duplicated genes in one string. In the VRPTW problem, the number of routes is not always the same in each individual, this leads to the difficulty of finding a good crossover operator. Inspired by the Uniform Order Crossover (UOX), Partially Mapped Crossover (PMX), and the Best Cost Route Crossover (BCRC) (Ombuki, Ross, & Han-shar, 2006), combined with our individual characteristics, a new but simple crossover operator is proposed which is named Best Sub-route Crossover (BSC). The main process of BSC can be described in 3 steps as follows:

Step 1: Two routes are selected from each parent randomly. As showed in Fig. 6, "route4" and "route2" are selected from the two parents respectively.

Step 2: In order to ensure that every node is unique for each individual, the duplicated nodes should be removed before the selected routes are joined to the other individuals. As showed in Fig. 7, the highlighted nodes are the duplicated ones in each parent.

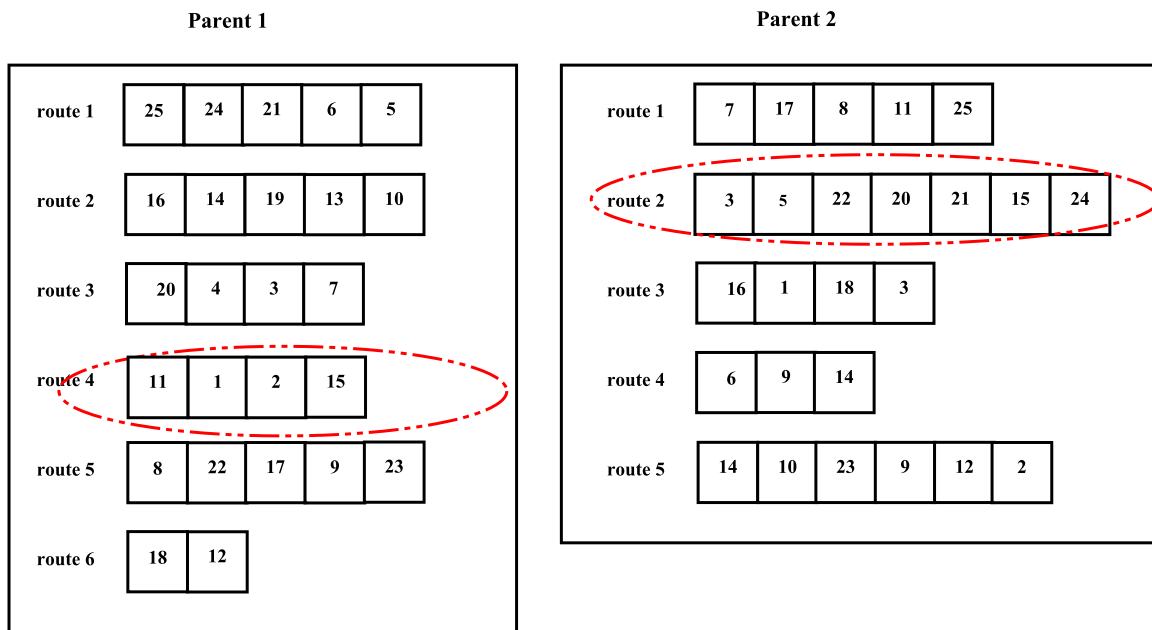


Fig. 6. The first step of the crossover operator.

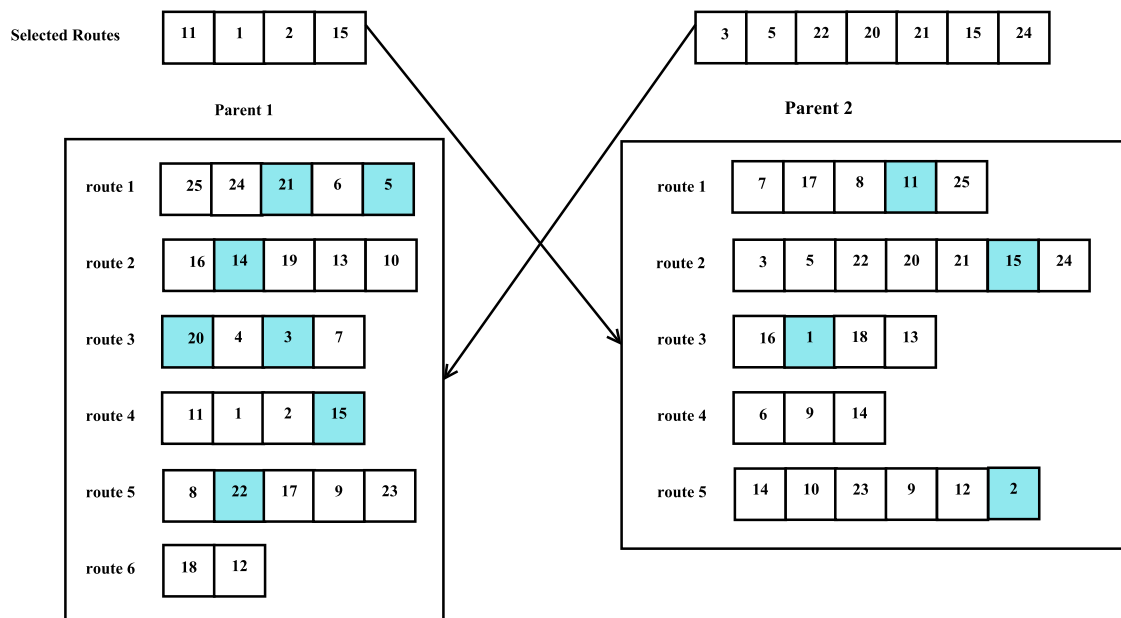


Fig. 7. The second step of the crossover operator.

Step 3: The duplicated nodes are removed, and the selected route from parent 1 is added to the parent 2, also the selected route from parent 2 is added to the parent 1. Only in this way can a route be exchanged completely, as well as ensuring that the offsprings are feasible individuals after the crossover. As showed in Fig. 8, the last routes in offspring 1 and offspring 2, namely “route7” and “route6” are added from the selected route in Step 1.

Because in the process of GA, there may exist some excellent routes in the individuals, by carrying out this simple operator, we can exchange the information between the routes, and ensure that the offspring is feasible at the same time. In the process of “Survival of the fittest”, the good routes can be maintained. It is evident that this operator results in an increased the number of routes

which seems to decrease the fitness value, while the local search operator can help us to improve the fitness value or even decrease the number of routes.

5.6. Mutation

Mutation is a genetic operator used to maintain genetic diversity from one generation of a population of genetic algorithm chromosomes to the next. It is known that the mutation operator is done in one individual. In the VRPTW, the classical single-point mutation is not suitable. Considering that the time window constraints are very strict, only the feasible individuals are allowed in the population. Given our chromosome structure, we used many mutation operators, they can be seen as follows:

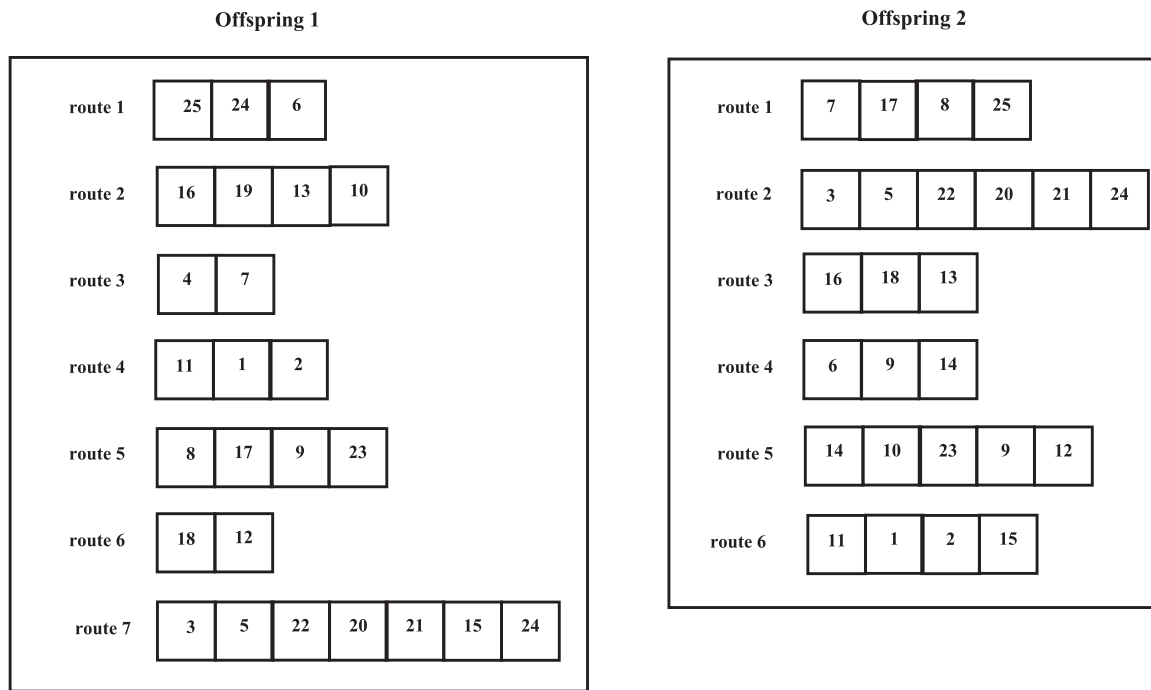


Fig. 8. The third step of the crossover operator.

(1) The operator can be undertaken in one route. We choose one route which has more than two nodes randomly and then undertake one of the operators:

- two-point swap: two nodes are selected randomly, and are swapped. If the route is still feasible, the new individual will replace the old one.
- reverse: a two-point cut will be selected randomly, and then reverse the cutting part. If the route is still feasible. The new individual will replace the old one.
- insert: one node is selected randomly and then deleted, insert this node in the route randomly again. If the route is still feasible, the new individual will replace the old one.

Note that the 3 operators are carried out randomly, and each individual only can use one operator in each generation.

(2) The mutation can also be carried out between two different routes. Two different routes will be randomly selected, and then one of the operators will be carried out.

- insert: a node will be selected randomly from one route, and the node will be inserted to another route. If a feasible position can be found, the new individual will replace the old one.
- part-exchange: two cutting places will be selected randomly, each cut can divide the routes into two pieces, then the two routes will swap the two parts, and combination two new routes. If the two new routes are still feasible, the new individual will replace the old one.

In order to improve the individuals, and enhance the convergence, the local search operators are introduced after the mutation operator. The detail of the operators can be described as follows.

λ -exchange is one of the best known neighbor search methods in heuristic algorithms. In this method, $\lambda = 2$. λ -exchange contains 3 kinds of operators, namely the 0-1 operator, 1-1 operator and 0-2 operator. The λ -exchange is carried out by randomly choosing two different routes in the same individual, and randomly choosing one of the operators. After undertaking the λ -exchange step, for any improvement of the solution, the offspring should be maintained, otherwise, the offspring should be dropped out. The different kinds of the λ -exchange operators can be described as follows:

- 0-1 operator is carried out by randomly picking up a patient from one route and finding the most suitable place to insert them into another route. If the insertion can improve the fitness value, the offspring will be maintained.
- 1-1 operator is carried out by randomly picking up a patient from these two routes respectively, and inserting each one into the other route. If the insertion can improve the fitness value, the offspring will be maintained.
- 0-2 operator is carried out by picking up two patients from one route, and finding the two best insert positions for them, and inserting these two nodes into each other's route. If the insertion can improve the fitness value, the offspring will be maintained.

5.7. Local search

Local search is a meta-heuristic method for solving computationally hard optimization problems. Local search can be used on problems that can be formulated by finding a solution that maximizes a criterion among a number of candidate solutions. Local search algorithms move from solution to solution in the space of candidate solutions (the search space) by applying local changes, until a solution deemed optimal is found or a time bound has elapsed.

Note that the local search operators differ from the mutation operator in two respects: (1) the aim of the local search operator is to produce an improvement in the solution, while the mutation is simply to make the population diversified which aims to avoid trapping in local optima in advance. (2) The mutation operator is executed just once in one iteration, while the local search operators are executed many times, until a solution deemed optimal is found or a time bound has elapsed.

6. Experimentation

To the best of our knowledge, no existing research has treated a similar model as an object of study, so there are no corresponding benchmarks to test the effectiveness of the proposed HGA. In order to check the effectiveness and efficiency of the heuristic algorithm proposed in this work, the fuzzy model is reduced into a classical Vehicle Routing Problem with Time Window (here, we assume fuzzy demand to be $d_{1,i} = d_{2,i} = d_{3,i}$, and the depot has the same location as the laboratory). Here we test the classical benchmark instances, to check the efficiency of the HGA.

6.1. Test on the classical VRPTW benchmark instances

The original Vehicle Routing Problems with Time Windows (VRPTW) instances were designed by Prof. Marius M. Solomon in 1983. According to the size of the instances, Solomon's instances can be divided into 3 versions: namely small sized instances with 25 customers, the medium sized instances with 50 customers, and the large sized instances with 100 customers. Based on the characteristics of the geographical distribution, these instances are divided into three categories: C-type (clustered customers), R-type (uniformly distributed customers) and RC-type (a mix of R and C types). Besides this, Homberger's instances were designed especially for the extra-large sized instances, which includes 200, 400, 600, 800 and 1000 customers. Solomon's benchmarks are very popularly tested in the research of VRPTW, while Homberger's benchmarks are especially used to test the for Large-Scare Vehicle Routing Problem with Time Window (LSVRPTW).

We have conducted our experiment on Intel Core i7-4770, 8 Duo 3.4 GHZ in order to solve the benchmarks. The benchmarks are selected from the Solomon and Homberger's instances. In small sized instances, the number of customers is 25, while in the large sized instances, the number of customers in each set is 100, and for the extra-large instances, the number of customers is 200.

For each instance, we have run the program for 15 times, then a statistics analysis is undertaken for the computing results. In order to describe the result clearly, some indicators are introduced to describe the results.

- (1) NO: the ID of the instance. According to common sense, for Solomon's instances, the ID consists of two parts. For example, the No.C101 shows the information: (a) "C" means the coordinate of the customers is distributed as a cluster distribution; (b) "C101" that means the instance comes from a data set named C101 (actually, C101 includes 3 different sizes of instances); Homberger's instances have the similar meaning of ID.
- (2) NC: the number of serviced customers.
- (3) NV: the number of the used vehicle.
- (4) avg. NV: the average number of the used vehicle.
- (4) best: the smallest objective value obtained in the 15 running times.
- (5) avg.: the mean objective value obtained in the 15 running times.
- (6) worst: the biggest objective value obtained in the 15 running times.
- (7) gap%: $(\text{best-TD})/\text{TD} * 100\%$
- (9) std: standard deviation value in the 15 running times.
- (10) std% = $\text{std}/\text{avg} * 100\%$.
- (11) CT: the computing time.
- (12) avg. CT: average computing time.
- (13) TD: Total Distance. Especially in the deterministic model, there is no additional distance, the total distance is equal to the planned distance.

Published best results for Solomon's Benchmarks (in Tables 3 and 4) are based on the tabu search, combination of SA, TA and GA

(Tan et al., 2001), localized genetic algorithm (Ursani et al., 2011), GA (Ombuki et al., 2006), decomposition technique and a genetic algorithm (Rousseau, Gendreau, Pesant, & Focacci, 2004). Published best results for Homberger's Benchmarks can be referred to paper (Mester & Bräysy, 2005) in Table 5.

As shown above, we can conclude that: (1) From gap% we find that the experimental results for most instances are very close to the best results, and this proves that the proposed HGA is quite efficient. (2) From the values of std%, we find that experimental results for most instances change slightly, and this reflects that the proposed HGA is robust.

Nonetheless, the aim of this paper is not to design a new and efficient algorithm to solve the Classical VRPTW, but to design an effective and efficient algorithm which can help us to solve the VRPTWFD. Additionally, in practical real life situations, most Home Health Care companies do not service more than 100 patients. Additionally, most researchers on routing problems with fuzzy demand do not use large scale instances (because the simulation process becomes time consuming for a sample size of 100 customers).

6.2. Test on the home health care routing problem with deterministic demand

In the classical VRPTW, each vehicle has to begin and end at the depot, while in VRPTWFD model, each vehicle has to begin from the depot and end at the laboratory. Because VRPTWFD is a typical fuzzy chance constraint programming problem, which could not be solved directly by the exact method, here we reduce the fuzzy demand to the deterministic demand, and compare with the results solved by HGA with the optimal solution to illustrate the efficiency of the proposed algorithm.

Because there are no instances for this problem, we obtain the instances by altering the Solomon's instances. We identify 5 parameters for the laboratory, namely: (a, b) is the location of the laboratory, e is the earliest visiting time of the laboratory, l is the latest visiting time of the laboratory, $[e, l]$ is the time window of the laboratory; w is the working time for the lab. In the experiment, 3 types of instances are used, namely: Random (R) type, Cluster (C) type, Random-Cluster (RC) type. For each instance, when a parameter changes, exact method and HGA are used to solve them. We use Gurobi solver to solve the models as the exact method.

Gurobi is a good commercial optimization solver for linear programming (LP), quadratic programming (QP), quadratically constrained programming (QCP), mixed integer linear programming (MILP), mixed-integer quadratic programming (MIQP), and mixed-integer quadratically constrained programming (MIQCP). Considering that the problem is NP-hard, only small scale instances can be solved in a limited time by Gurobi. Here we have only used 25-customers instances. The experimental results can be found in Table 6.

The experimental results show that the HGA and Gurobi almost obtain the same solution, so it is clear that the HGA performs well when solving the model with deterministic demand.

6.3. Test on the home health care routing problem with fuzzy demand

Now, we will offer some instances to illustrate the fuzzy demand approach, and show how the hybrid heuristic algorithm works. Two types of experiment conditions are created based on the size of the problem. (We give two instances, the small-sized instance with 25 patients, and the large-sized instance with 100 patients, which are detailed in Table 7)

The value of the dispatcher preference index DPI varies within the interval $[0, 1]$ with a step of 0.1.

Table 3

The experimental results for the small size of Solomon's instances with the CVRPTW model.

Hybrid genetic algorithm										Published best results	
NO.	NC	avg.NV	best	gap1%	avg.	worst	std	std%	CT	NV	TD
C101	25	3.00	191.81	0.00	191.81	191.81	0.00	0.00	34.55	3	191.81
C102	25	3.00	190.74	0.00	190.78	191.13	0.12	0.06	35.15	3	190.74
C103	25	3.00	190.74	0.00	191.24	194.54	1.18	0.62	33.93	3	190.74
C104	25	3.00	187.45	0.00	192.88	224.45	11.39	5.91	30.72	3	187.45
C105	25	3.00	191.81	0.00	191.81	191.81	0.00	0.00	31.40	3	191.81
C106	25	3.00	191.81	0.00	191.81	191.81	0.00	0.00	34.42	3	191.81
C107	25	3.00	191.81	0.00	191.81	191.81	0.00	0.00	31.05	3	191.81
C108	25	3.00	191.81	0.00	191.81	191.81	0.00	0.00	29.86	3	191.81
C109	25	3.00	191.81	0.00	195.85	232.16	12.76	6.51	31.53	3	191.81
R101	25	8.10	618.33	0.00	624.45	639.93	8.46	1.36	18.24	8	618.33
R102	25	7.00	548.11	0.13	549.98	564.30	5.05	0.92	19.06	7	547.40
R103	25	5.60	464.83	2.00	468.49	473.92	4.24	0.90	21.12	6	455.70
R104	25	4.50	417.96	0.00	428.89	446.13	11.83	2.76	23.74	4	417.96
R105	25	6.00	531.54	0.00	533.06	538.36	2.79	0.52	19.66	6	531.54
R106	25	5.00	466.48	0.00	473.07	483.57	8.02	1.70	21.83	5	466.48
R107	25	4.60	425.27	0.00	432.77	440.74	7.31	1.69	23.32	4	425.27
R108	25	4.00	398.29	0.00	405.53	436.12	11.74	2.90	24.66	4	398.30
R109	25	5.00	442.63	0.00	451.31	464.38	8.45	1.87	21.41	5	442.63
R110	25	4.90	445.18	0.00	451.61	454.97	3.26	0.72	21.60	5	445.18
R111	25	5.00	431.12	0.33	438.31	444.74	5.30	1.21	22.20	4	429.70
R112	25	4.00	394.10	0.00	408.09	425.43	10.87	2.66	23.48	4	394.10
RC101	25	4.00	462.16	0.23	462.16	462.16	0.00	0.00	23.64	4	461.10
RC102	25	3.00	352.74	0.00	352.74	352.74	0.00	0.00	30.59	3	352.74
RC103	25	3.00	333.92	0.00	334.05	334.18	0.14	0.04	30.81	3	333.92
RC104	25	3.00	308.39	0.41	309.03	309.22	0.32	0.10	30.62	3	307.14
RC105	25	4.00	412.38	0.00	412.38	412.38	0.00	0.00	24.28	4	412.38
RC106	25	3.00	346.51	0.00	346.51	346.51	0.00	0.00	30.52	3	346.50
RC107	25	3.00	298.95	0.00	298.95	298.95	0.00	0.00	30.50	3	298.95
RC108	25	3.00	294.99	0.00	295.74	296.19	0.47	0.16	31.12	3	294.99
C201	25	2.00	215.54	0.39	215.54	215.54	0.00	0.00	45.56	2	214.7
C202	25	1.20	215.54	0.39	226.24	232.91	6.42	2.76	14.51	2	214.7
C204	25	1.80	215.37	1.07	229.00	241.40	9.88	4.09	25.13	2	213.1
C205	25	2.00	215.54	0.39	215.54	215.54	0.00	0.00	34.07	2	214.7
C206	25	2.00	215.54	0.39	215.54	215.54	0.00	0.00	33.35	2	214.7
C207	25	2.00	215.34	0.30	215.60	216.25	0.38	0.17	26.16	2	214.7
R201	25	4.20	464.37	0.23	473.72	487.27	10.07	2.07	18.24	4	463.3
R202	25	3.80	411.49	0.24	413.05	418.59	3.11	0.74	20.03	4	410.5
R203	25	3.00	394.70	0.84	402.06	424.21	12.78	3.01	24.98	3	391.4
R204	25	2.40	355.89	0.25	362.59	373.55	8.18	2.19	23.64	2	355
R205	25	2.80	395.82	0.72	405.62	417.77	8.10	1.94	20.67	3	393
R206	25	1.80	379.85	1.45	415.25	467.19	46.35	9.92	23.48	3	374.4
R207	25	1.80	369.21	2.10	399.43	435.12	30.52	7.01	17.81	3	361.6
R209	25	2.60	373.08	0.64	379.45	390.49	7.44	1.91	21.02	2	370.7
R210	25	2.20	408.31	0.92	413.05	416.77	3.11	0.75	27.49	3	404.6
RC201	25	3.00	361.24	0.29	361.24	361.24	0.00	0.00	25.06	3	360.2
RC202	25	3.00	338.82	0.24	339.42	339.57	0.34	0.10	25.13	3	338
RC203	25	2.40	330.02	0.96	359.03	437.92	45.33	10.35	15.76	3	326.9
RC205	25	3.00	338.93	0.27	338.93	338.93	0.00	0.00	22.63	3	338
RC206	25	2.40	326.14	0.66	347.26	373.39	21.83	5.85	24.30	3	324
RC207	25	2.60	298.95	0.22	314.00	345.51	18.92	5.48	29.18	3	298.3

The results for the instance 1 can be found in Table 8, Figs. 9, and 10. The results for the instance 2 can be found in Table 9, Figs. 11, and 12. These tables and figures show tendencies regarding planned distances, additional distances due to failures, the total distance and other indicators that vehicles covered as the dispatcher preference index varies.

We can see that with the increase of the *DPI* value, the planned distance tends to increase (but not very strictly), while the additional distance decreases and there is no clear trend in the total distance.

For instance 1, the additional distance strictly decreases as *DPI* value increases from 0 to 0.8. However, when $DPI \in [0.8, 1]$, the additional distance becomes 0, and this means that there is no failure route. From this result, we can easily conclude that 0.8 is the best *DPI* value to make the planed route.

For instance 2, the additional distance strictly decreases as *DPI* value increases from 0 to 0.8. However, when $DPI \in [0.8, 1]$, the

additional distance becomes 0, that means there is no failure route. From this result, we can easily conclude that 0.8 is the best *DPI* value to make the planned route.

As we have discussed previously, when the *DPI* value is very low, the decision-maker subjectively desires to make full use of the vehicle, so the planned distance is very small. In this situation, the chance that the vehicles may not meet the patients' demand is increased, and the nurse has to return to the depot to load more drugs. These failure routes bring more additional distances, which also leads to the delay of the service time. Higher values of *DPI* are characterized by lower utilization of vehicle capacity along the planned routes and less additional distance due to fewer failures.

Moreover, from Figs. 10 and 12, it is clear that indicators like Percentage of Delay in service time, Average Delayed Time are at 0, while the NV increases, when the *DPI* increases. This tendency shows that the higher *DPI* value can help us provide a better level of service within the service time window.

Table 4

The experimental results for the big size of Solomon's instances with the CVRPTW model.

NO.	NC	Hybrid genetic algorithm								Published best results	
		Avg.NV	best	gap%	avg.	worst	std	std%	CT	NV	TD
C101	100	10.70	828.94	0.00	858.11	937.06	33.92	3.95	148.10	10	828.94
C102	100	10.80	828.94	0.00	888.55	946.51	34.59	3.89	147.77	10	828.94
C103	100	10.90	831.36	0.40	909.99	1,000.87	51.69	5.68	149.64	10	828.06
C104	100	11.00	829.63	0.59	919.92	1,016.06	40.91	4.45	152.40	10	824.78
C105	100	11.20	828.94	0.00	897.55	997.25	51.45	5.73	146.33	10	828.94
C106	100	10.70	828.94	0.00	882.89	1,011.28	64.82	7.34	147.58	10	828.94
C107	100	11.10	828.94	0.00	883.47	930.47	38.93	4.41	145.97	10	828.94
C108	100	11.10	828.94	0.00	902.17	1,033.27	57.65	6.39	148.34	10	828.94
C109	100	11.50	830.91	0.24	959.95	1,075.61	69.68	7.26	151.80	10	828.94
R101	100	20.60	1,651.48	0.52	1,680.36	1,695.53	14.56	0.87	127.90	20	1,642.87
R102	100	18.70	1,477.41	0.35	1,512.01	1,529.45	18.70	1.24	134.37	18	1,472.28
R103	100	15.70	1,259.47	3.78	1,285.36	1,312.43	19.37	1.51	146.04	14	1,213.62
R104	100	12.00	1,046.64	3.90	1,081.09	1,127.53	23.06	2.13	168.88	10	1,007.31
R105	100	15.80	1,405.96	3.32	1,428.68	1,458.41	16.28	1.14	143.09	15	1,360.78
R106	100	13.80	1,279.19	3.03	1,312.13	1,398.31	36.19	2.76	155.53	13	1,241.52
R107	100	12.50	1,134.91	2.74	1,162.48	1,196.99	21.46	1.85	164.07	11	1,104.66
R108	100	11.30	998.60	3.93	1,026.44	1,056.92	20.32	1.98	172.45	10	960.88
R109	100	13.50	1,197.63	3.98	1,242.57	1,280.46	27.81	2.24	150.50	13	1,151.84
RC101	100	16.90	1,699.22	0.13	1,722.76	1,766.47	19.76	1.15	98.07	15	1,696.94
RC102	100	14.90	1,501.39	2.36	1,531.43	1,563.90	20.64	1.35	105.29	14	1,466.84
RC103	100	12.90	1,325.01	5.02	1,365.53	1,426.44	29.33	2.15	114.75	11	1,261.67
RC104	100	11.60	1,178.44	3.78	1,230.78	1,284.56	41.85	3.40	118.81	10	1,135.48
RC105	100	16.70	1,560.57	2.76	1,621.51	1,668.86	38.30	2.36	102.10	16	1,518.60
RC106	100	14.20	1,435.13	0.73	1,459.28	1,475.89	12.02	0.82	108.23	13	1,424.73
C201	100	3.00	591.56	0.00	591.56	591.56	0.00	0.00	349.76	3.00	591.56

Table 5

The experimental results for Homberger's instances with the CVRPTW model.

NO.	NC	Hybrid genetic algorithm								Published best results	
		avg.NV	best	gap%	avg.	worst	std	std%	CT	NV	TD
C1_2_1	200	20.2	2704.58	0.00	2736.99	2862.34	70.0963	2.56	552.29	20	2704.58
C1_2_2	200	20.8	2949.46	0.00	2825.88	2895.20	71.1551	2.52	551.40	18	2949.46
C1_2_3	200	20.4	2726.68	0.69	2812.13	2955.65	89.5413	3.18	541.11	18	2708.08
C1_2_4	200	20.2	2705.05	2.29	2830.97	2978.66	108.4239	3.83	546.07	18	2644.61
C1_2_5	200	21.6	2777.42	2.79	2883.82	2989.30	75.578	2.62	541.19	20	2702.05
C1_2_6	200	22.6	2856.47	5.75	2964.74	3149.93	124.1854	4.19	532.76	20	2701.04
C1_2_7	200	21	2701.04	0.00	2788.52	2927.45	89.1981	3.20	536.55	20	2701.04
C1_2_8	200	21.2	2816.71	1.72	2898.20	3011.03	74.9556	2.59	539.53	18	2769.19
C1_2_9	200	20.4	2682.55	1.50	2812.50	2919.55	100.6867	3.58	548.33	18	2642.82
R1_2_1	200	24	4827.59	1.39	4939.49	5015.04	71.8221	1.45	526.97	20	4761.61
R1_2_2	200	22	4102.42	1.18	4171.95	4243.16	57.034	1.37	530.19	18	4054.44
R1_2_4	200	16.6	3054.03	-0.45	3135.22	3242.09	77.1924	2.46	505.37	18	3067.93
R1_2_5	200	20.6	4200.42	2.13	4262.75	4327.12	52.6876	1.24	509.61	18	4112.88
RC1_2_1	200	21.6	3644.77	-1.28	3708.13	3773.60	60.7483	1.64	518.43	18	3691.99
RC1_2_2	200	20.2	3374.56	2.30	3432.90	3494.63	53.079	1.55	530.50	18	3298.68
RC1_2_3	200	19.2	3159.20	4.41	3235.68	3306.20	56.7889	1.76	528.49	18	3025.90
RC1_2_4	200	17.2	2772.14	-3.73	2853.84	2936.19	64.9983	2.28	519.39	18	2879.40
RC1_2_5	200	19.8	3455.62	1.05	3553.35	3659.70	84.6549	2.38	523.21	18	3419.81
RC1_2_6	200	19.8	3424.69	0.93	3519.21	3630.45	81.5722	2.32	522.35	18	3393.09
RC1_2_7	200	19.4	3315.18	1.49	3374.29	3420.39	51.6562	1.53	515.56	18	3266.48
RC1_2_8	200	19	3190.68	2.40	3276.31	3361.82	72.4416	2.21	537.07	18	3115.82
RC1_2_9	200	18.2	3119.28	2.65	3189.60	3298.58	72.0995	2.26	513.05	18	3038.85

Table 6

The experimental results for the small size instances of the model with deterministic demand.

No.	HGA			Exact method			GAP	Parameter of the lab
	NV	TD	CT(s)	NV	TD	CT(s)		
R101	8	618.3299	2.48	8	618.3299	0.323	0.00%	$a = 35; b = 35; e = 0; l = 230; w = 0;$
R101-1	8	650.4592	4.58	8	650.4592	10.386	0.00%	$a = 50; b = 50; e = 0; l = 230; w = 0;$
R101-2	8	670.7533	8.05	8	670.7533	8.226	0.00%	$a = 40; b = 60; e = 0; l = 230; w = 0;$
R101-3	8	618.3299	2.48	8	618.3299	0.323	0.00%	$a = 35; b = 35; e = 50; l = 230; w = 0;$
R101-4	8	618.3299	2.48	8	618.3299	0.323	0.00%	$a = 35; b = 35; e = 0; l = 250; w = 0;$
C101	3	191.8136	1.06	3	191.8136	0.506	0.00%	$a = 40; b = 50; e = 0; l = 1236; w = 0;$
C101-1	3	212.0759	1.96	3	212.0759	0.423	0.00%	$a = 10; b = 80; e = 0; l = 1236; w = 0;$
C101-2	3	308.5431	1.74	3	308.5431	1.425	0.00%	$a = 80; b = 30; e = 0; l = 1236; w = 0;$
C101-3	3	191.8136	1.06	3	191.8136	0.506	0.00%	$a = 40; b = 50; e = 50; l = 1236; w = 0;$
C101-4	3	191.8136	1.06	3	191.8136	0.506	0.00%	$a = 40; b = 50; e = 50; l = 1200; w = 0;$

Table 7

The relative values of the test instances.

ID of instances	No. of patients	No. of the available vehicles	Vehicle capacity	Coordinate of the depot	Coordinate of the laboratory	Notes
Instance 1	25	20	150	(45, 50)	(30, 60)	Small size
Instance 2	100	20	150	(40, 50)	(45, 45)	Big size

Table 8

The computing results for instance 1 with different DPI.

DPI	NV	TD	PD	AD	TDT	PDP	ADT
0	3	295.15	175.35	119.8	495.48	56.24%	35.248
0.1	3	358.56	231.6	126.97	574.27	59.14%	38.591
0.2	3	354.26	231.6	122.66	565.63	58.40%	38.457
0.3	3	361.23	222.87	138.36	403.22	58.26%	27.478
0.4	4	308.11	243.94	64.166	98.624	5.48%	60.869
0.5	4	295	267.42	27.578	64.353	10.74%	21.414
0.6	4	273.06	252.2	20.861	37.878	7.28%	12.265
0.7	5	263.76	259.98	3.7733	4.8294	0.56%	3.7048
0.8	5	264.46	264.46	0	0	0.00%	0
0.9	6	323.67	323.67	0	0	0.00%	0
1	6	320.44	320.44	0	0	0.00%	0

Table 9

The computing results for instance 2 with different DPI.

DPI	NV	TD	PD	AD	TDT	PDP	ADT
0	10	1431.1	838.82	592.3	2496	61.86%	40.353
0.1	13	1588.6	1137.2	451.41	1299.7	33.75%	38.559
0.2	14	1475.5	1203.3	272.21	1000.1	34.33%	29.153
0.3	14	1578.9	1263.6	315.37	954.07	32.11%	29.705
0.4	14	1480.9	1265.1	215.8	403.52	10.66%	38.701
0.5	14	1544.9	1263.2	281.67	685.32	29.21%	23.422
0.6	15	1625	1541.6	83.364	159.39	5.85%	27.301
0.7	16	1422.6	1419	3.6043	4.5935	0.13%	3.3798
0.8	17	1489.4	1489.4	0	0	0.00%	0
0.9	19	1489.9	1489.9	0	0	0.00%	0
1	19	1499.4	1499.4	0	0	0.00%	0

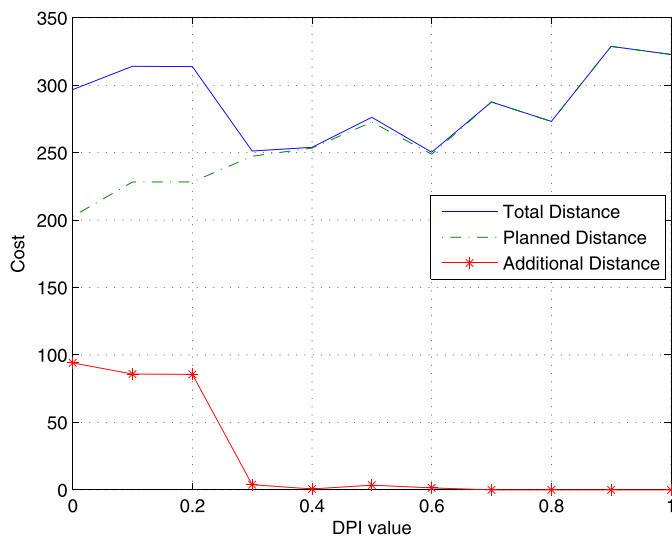
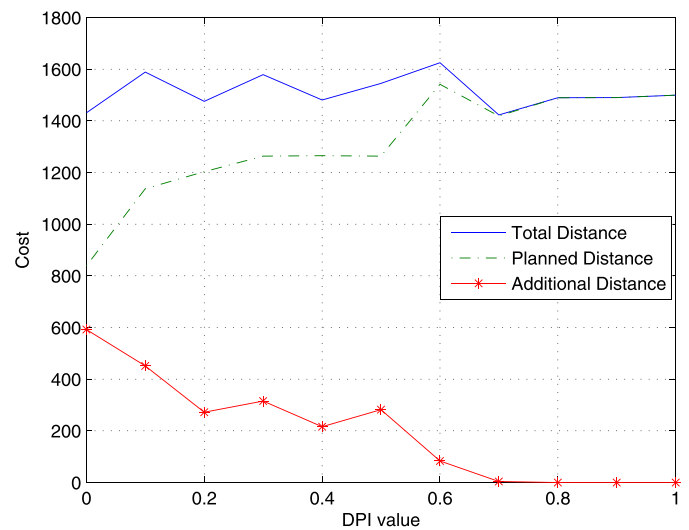
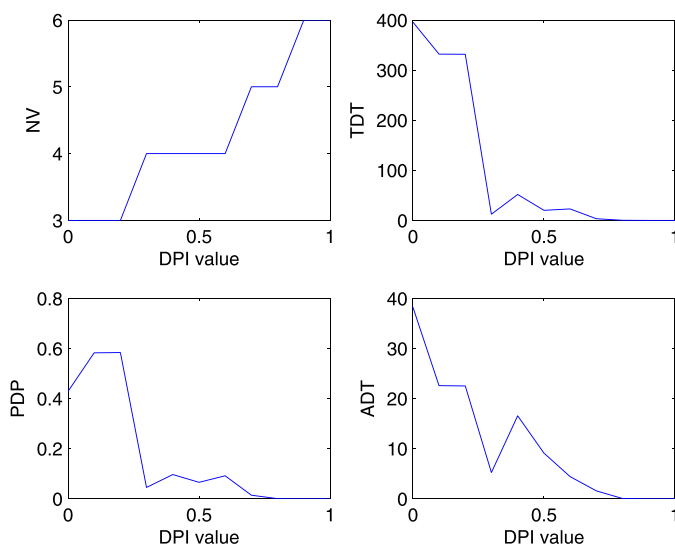
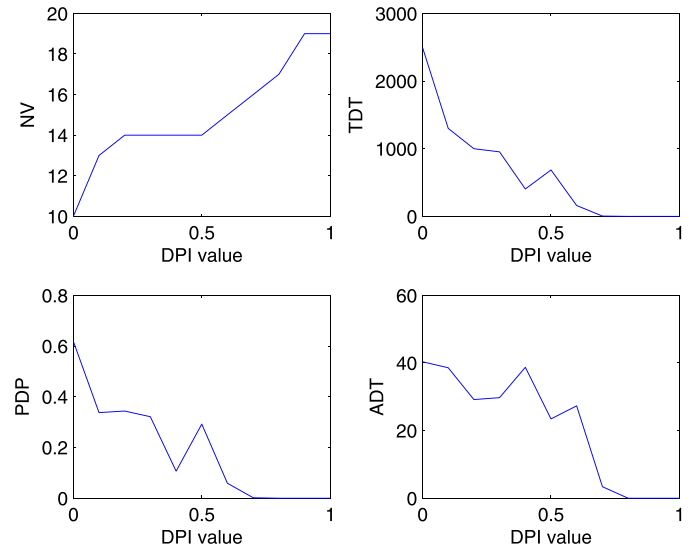
**Fig. 9.** Cost changes for different DPI values for instance 1.**Fig. 11.** Cost changes for different DPI values for instance 2.**Fig. 10.** Indicators changes for different DPI values for instance 1.**Fig. 12.** Indicators change for the different DPI values for instance 2.

Table 10

The comparison between the best solution obtained from the deterministic model and fuzzy model.

Instance no.	Solution from	NV	TD	PD	AD	TDI	PDP	ADT
Instance 1	deterministic model	4	355.88	235.38	120.51	335.17	40.58%	33.054
	fuzzy model	5	264.46	264.46	0	0	0.00%	0
Instance 2	deterministic model	15	1719.0	1142.7	576.3192	1855.2	39.73%	46.789
	fuzzy model	17	1489.4	1489.4	0	0	0.00%	0

Table 11

Comparison between the solutions obtained by deterministic model and fuzzy model.

Instance Name	Solutions obtained by deterministic model									Solutions obtained by fuzzy model								
	NV	TD	PD	AD	TDI	PDP	ADT	CT		DPI	NV	TD	PD	AD	TDI	PDP	ADT	CT
HHC_1	4	355.88	235.38	120.51	335.17	40.58%	33.05	15.00		0.80	5	264.46	264.46	0.00	0.00	0.00%	0.00	178.00
HHC_2	4	451.00	335.69	115.31	7876.99	57.35%	550.67	16.08		0.60	5	385.22	384.15	1.07	19.16	0.84%	3.45	177.90
HHC_3	4	438.01	318.24	119.77	644.51	48.13%	53.56	15.54		0.30	4	373.21	332.80	40.40	351.46	28.08%	50.07	169.78
HHC_4	4	442.03	330.58	111.45	428.33	33.70%	50.69	15.42		0.50	4	375.88	339.03	36.86	198.92	8.96%	54.48	175.67
HHC_5	8	634.78	634.78	0.00	0.00	0.00%	0.00	12.96		0.30	8	634.78	634.78	0.00	0.00	0.00%	0.00	183.55
HHC_6	8	583.37	583.37	0.00	0.00	0.00%	0.00	13.18		0.30	7	571.01	571.01	0.00	0.00	0.00%	0.00	166.53
HHC_7	5	498.99	498.99	0.00	0.00	0.00%	0.00	13.78		0.20	5	478.37	478.37	0.00	0.00	0.00%	0.00	177.29
HHC_8	5	429.11	429.11	0.00	0.00	0.00%	0.00	14.03		1.00	5	437.74	437.74	0.00	0.00	0.00%	0.00	175.70
HHC_9	7	605.79	605.79	0.00	0.00	0.00%	0.00	14.27		0.30	6	537.47	537.47	0.00	0.00	0.00%	0.00	174.25
HHC_10	7	525.56	525.56	0.00	0.00	0.00%	0.00	13.34		0.50	6	496.77	496.77	0.00	0.00	0.00%	0.00	174.05
HHC_11	5	484.40	484.40	0.00	0.00	0.00%	0.00	14.22		0.60	5	437.50	437.50	0.00	0.00	0.00%	0.00	171.70
HHC_12	5	446.20	446.20	0.00	0.00	0.00%	0.00	13.85		0.10	4	415.88	415.88	0.00	0.00	0.00%	0.00	169.99
HHC_13	7	536.29	536.29	0.00	0.00	0.00%	0.00	13.69		0.40	5	470.61	470.61	0.00	0.00	0.00%	0.00	169.08
HHC_14	5	501.38	500.04	1.34	30.99	1.68%	4.43	14.08		0.80	5	452.79	452.79	0.00	0.00	0.00%	0.00	178.56
HHC_15	5	475.62	475.10	0.51	2.31	0.10%	0.77	13.91		0.90	6	473.13	473.13	0.00	0.00	0.00%	0.00	175.76
HHC_16	4	448.47	447.86	0.61	8.72	0.29%	1.45	15.10		0.70	4	412.16	411.82	0.34	6.70	0.30%	1.34	170.33
HHC_17	5	581.19	486.68	94.51	168.26	15.57%	34.05	13.97		0.80	6	505.97	504.59	1.38	2.52	0.18%	1.11	170.69
HHC_18	5	550.06	448.69	101.37	298.55	26.46%	39.70	13.97		0.60	6	488.28	486.87	1.40	2.23	0.16%	1.12	169.42
HHC_19	5	585.21	441.26	143.95	1144.94	45.26%	101.54	14.17		0.20	5	480.20	409.03	71.17	422.81	24.00%	70.47	182.83
HHC_20	4	577.28	406.29	170.99	1093.20	46.23%	94.91	14.86		0.80	6	463.69	462.99	0.70	1.82	0.12%	0.61	181.34
HHC_21	6	528.29	481.44	46.85	113.16	5.67%	39.17	14.74		0.30	6	494.53	493.08	1.45	2.80	0.18%	1.25	182.17
HHC_22	5	548.74	431.26	117.47	387.83	28.03%	50.81	14.09		0.80	6	479.68	476.95	2.74	5.75	0.38%	2.45	177.58
HHC_23	5	577.86	416.98	160.88	941.39	45.92%	81.90	14.92		0.20	6	475.87	475.17	0.70	1.44	0.12%	0.48	174.63
HHC_24	5	606.39	412.23	194.16	1369.44	57.97%	95.61	13.75		0.10	6	468.18	468.18	0.00	0.00	0.00%	0.00	169.93

Table 12

Friedman-test on the indicators obtained by two different models.

Model	Deterministic model	Fuzzy model	Deterministic model	Fuzzy model	Deterministic model	Fuzzy model
Indicator	TD	TD	AD	AD	PDP	PDP
Avg. ranking	1.92	1.04	1.625	1	1.583	1.04
P-value	1.19E-05		0.0001		0.0008	

6.4. Comparison between the solutions obtained by two models

In the last part of this section, we will illustrate the strength of the model with fuzzy demand. For each instance (namely instance 1 and instance 2 in Table 7), we firstly solve the model with deterministic demand by HGA, obtain the best vehicle scheduling solution, then employ the stochastic simulation method to simulate the scheduling indicators while demand changes. These results for the indicators can be found in Table 10.

By comparing Tables 8–10, we observe that if we choose the best DPI value obtained by the model with fuzzy demand to plan the route, this solution works much better than the best solution obtained by the model with deterministic demand. Table 10 also shows that the solution obtained by the model with deterministic demand leads more easily to failure routes and delayed service while demand is not stable. These results illustrate that the model with fuzzy demand is more flexible and robust. Considering that uncertain demand is more common in real life, the proposed VRPTWFD is meaningful.

In Table 10, we have carried out a comparison between the solutions obtained by the fuzzy model and deterministic model to highlight the fuzzy model. However, we just compared two instances, which is not sufficient to validate the strength of the fuzzy model. We will complement more comparisons, and employ a statistical test to validate the strength.

We create some new instances from altering the famous Solomon's 25-customer instances for VRPTW. Our instances differ from the Solomon's instances in the following respects:

- (1) The capacity of each vehicle becomes 150;
- (2) We add a laboratory point which is not in the same place with the depot.
- (3) We assume that the fuzzy demand for each patient is $\tilde{d}_i = (0.8 * d_i, d_i, 1.8 * d_i)$.

The comparison results are showed in Table 11, it illustrates the significant of the GAP between the stochastic model and deterministic model.

There are many instances available in the literature, however, due to the restriction of the space and time, just 24 instances are considered in this section. To determinate which solution is better, Friedman-test is performed on the results.

The Friedman-test is a non-parametric statistical test which is proposed by Milton Friedman. This method aims to detect differences in treatments across multiple test attempts. Considering the specific characteristics of our work, we detect two different method: namely the deterministic model and the fuzzy model. Two hypothesis are proposed in the following.

H_0 : difference between the results obtained by two models follows a symmetric distribution around zero;

H_1 : difference between the results obtained by the two models does not follow a symmetric distribution around zero.

We have performed 3 Friedman-test on 3 indicators obtained by two models respectively, then we get p -value, and the average ranking value (Avg. ranking) for each indicator, the specific results are showed in Table 12.

The p -value for Friedman-test on the TD is 1.19×10^{-5} (less than 0.5, reject the H_0). The p -value for Friedman-test on the AD is 0.0001 (less than 0.5, reject the H_0). The p -value for Friedman-test on the PDP is 0.0008 (less than 0.5, reject the H_0). We conclude the following two points:

- (1) Each p -value indicates the significance differences between the model within and without considering fuzzy demand.
- (2) From the Avg. ranking values, it illustrates that total distance (TD), additional distance (AD) and percentage of the delayed patients (PDP) would be less if we take into account the fuzzy demand when we make a decision on the Home health Care routing. So, the decision of arranging and assigning routes would be more reasonable if we take into account the fuzzy demand.

However, according to the results in Table 11, we can easily find that, the fuzzy model nearly takes 10 times the computing cost than the deterministic model. So, in practical, when the quantity of demand is deterministic, we may not recommend to employ the fuzzy model. Additionally, from the results of HHC_5-HHC_13, the solutions obtained by the fuzzy model seem not have a big improvement than the solutions obtained by the deterministic model, because in these instances the capacity of each vehicle is quite sufficient for the demands of patients. So, when the capacity of vehicle is quite sufficient for the demand (for example, the capacity is 1000, and the demand is very small), we may neglect the capacity constraints, nor does consider fuzzy demand.

7. Conclusion

Since transportation costs constitute one of the largest forms of expenditure in the Home Health Care industry, it is of great significance to research the optimization of the Home Health Care logistics. Based on a survey of the Home Health Care companies, the basic operational process illustrates that the demand for the required drugs for each patient is non-deterministic when the HHC company makes a decision on the scheduling. This paper assumes the uncertain demand as a fuzzy variable, which is closer to real life. The vehicle routing problem with time window and fuzzy demand (VRPTWFD) is considered, and a fuzzy chance constraint is constructed based on the fuzzy credibility theory. According to related research work, the stochastic simulation method and hybrid genetic algorithm are integrated to solve the proposed model.

We have conducted 3 series of experiments to test the HGA.

Firstly, VRPTWFD has been reduced to the classical VRPTW, and HGA has been applied to solve the Solomon's instances. A comparison between the results obtained by HGA and the published best results shows that HGA performs well in both large and small size instances.

Secondly, since VRPTWFD is a fuzzy chance constraint programming problem and could not be solved by directly by an exact method, we firstly reduced the fuzzy demand to the deterministic demand. By comparing the results obtained by both HGA and Gurobi Solver, we found that HGA is efficient in solving the model with a deterministic demand.

Thirdly, two different sizes of test problems were generated to test the model with fuzzy demand and the computational experiments showed that the dispatcher preference index (DPI) greatly influenced the planned routes, additional distances, and service delay. The best value of DPI was obtained via conducting computational experiments for different values of this parameter.

This paper provides a potential basis for future research. On the one hand, we can continue to consider other non-deterministic parameters such as the stochastic traveling time, which brings the model closer to the HHC practical activities. On the other hand, we may also try to design other algorithms such as Lagrange-relaxation decomposition and column generation to estimate the lower bound of our model or even obtain the optimal solution.

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